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To fear or not? The impact of internet use on Chinese public safety perception

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ABSTRACT

Drawing on nationally representative data from the 2017 and 2021 Chinese Social Survey, this paper examines how Internet use shapes public safety perception (PSP) in China. Across multiple linear regressions, logit models, multilevel analyses, and instrumental-variable (2SLS) estimates, we find a robust negative relationship: greater Internet use decreases people's PSP. The effect, however, is heterogeneous. By purpose, social-information use consistently eroded PSP, while the relationship between entertainment-oriented use and PSP exhibited a cross-wave divergence which appeared neutral in 2017 but positive in 2021, reflecting the buffering and diversionary role of online entertainment during the pandemic. By perception type, the Internet has a stronger depressive effect on societal-level evaluations of safety than on individual-level assessments. Subgroup analyses show sharper declines among women and the less educated, and provincial context was more salient in 2021 the wave, with higher average education predicting lower PSP. These findings underscore how digital exposure interacts with usage patterns and social environments, carrying important implications for governance and risk communication.

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Internet use; public safety perception; China; Chinese Social Survey (CSS); empirical analysis

1. Introduction

Since Ulrich Beck introduced the influential concept of the 'risk society', awareness of systemic crises and the imperative of risk prevention have become central concerns in contemporary public governance (Rosa, Renn, and McCright 2014). A 'risk society' refers to a modern social paradigm in which manufactured risks—such as climate change, pandemics, and technological disasters—increasingly overshadow natural hazards. Unlike pre-industrial dangers, these anthropogenic risks are often invisible, statistically assessed, and paradoxically produced by the very systems intended to promote progress and security. Within this context, public safety perception (PSP) has gradually risen to prominence on the public agenda, becoming a focal point of current academic inquiry. Scholars regard PSP as both a psychological foundation for understanding individuals' protective behaviors (Slavich 2020) and a sociopolitical backdrop for interpreting

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governmental security policies (Lucchesi et al. 2022). Existing research has examined a wide range of factors influencing these perceptions, encompassing objective elements such as living environments (Nielsen and Smyth 2008) and sociodemographic characteristics (Li, Chen, and Zhu 2021), as well as subjective aspects including personal identity (He et al. 2020), values (Torres et al. 2023), and trust (Eriksson 2011). Collectively, these studies provide important insights into how PSP are formed and understood within a risk society.

However, research on the determinants of PSPs remains limited in two respects. First, the rapid expansion of the Internet has introduced a new and increasingly important influence on these perceptions (Wang et al. 2022; Yi 2024). As digital connectivity becomes deeply embedded in daily life (Long and Yi 2019), public sensitivity to safety risks has risen alongside growing Internet dependence. This dual trend has prompted inquiries into their relationship (Twumasi et al. 2021; Zheng, Fan, and Jia 2022), yet existing studies offer insufficient insight—particularly in national contexts where Internet use is pervasive. Second, the Chinese case remains underexplored despite its global significance. The government has promoted reforms aimed at building a 'safety society' (Huo and Lin 2022), while China's vast digital landscape—1.1 billion Internet users, most accessing *via* mobile phones—makes it an especially important case for examining how Internet use shapes PSP (CNNIC 2024). Understanding how such widespread Internet use influences PSP in China not only addresses a key empirical gap but also offers valuable insights for the broader study of safety perceptions in the digital age.

To address this gap, this study uses data from the 2017 and 2021 waves of the Chinese Social Survey (CSS) to examine how Internet use shapes PSP. These waves offer recent, comparable data with consistent questionnaire items, enabling analysis across time. Because the COVID-19 pandemic affected both Internet access (Zhao et al. 2020) and perceptions of safety (Chen et al. 2021), comparing 2017 and 2021 provides a unique opportunity to examine how the association between Internet use and PSP manifested differently before and after the pandemic. Multivariate regression and instrumental variable estimates reveal that Internet use generally lowers individuals' sense of public safety. However, the magnitude and even the direction of this effect differ by usage purpose—social-information use is negatively associated with public safety perception in both waves, whereas entertainment use is positively associated with PSP only in the 2021 wave. In addition, the study also explores heterogeneous and contextual patterns, showing how the relationship between Internet use and public safety perception varies across social groups and regional environments. These findings advance understanding of the nuanced ways Internet use influences PSP in contemporary China.

2. Theoretical framework

2.1. Public safety perception

As concerns over safety intensify, PSP has attracted growing scholarly attention due to their significant social relevance. For individuals, these perceptions may align with or diverge from actual safety conditions. Regardless of this alignment, they influence protective behaviors (Quansah et al. 2022) and shape how people evaluate and support governmental safety policies (Tan et al. 2022).

The existing literature identifies a range of factors influencing PSP, including actual security conditions as well as demographic, social, and psychological variables (Mulvey 2002; Weimann et al. 2017). Demographically, men often report lower safety perceptions than women (Willits, Theodori, and Luloff 2016; Yu et al. 2018), and non-Whites express greater concern over environmental safety than Whites (Olawoyin et al. 2016). Older adults also tend to report lower PSP than younger adults (Brasier et al. 2013). Social influences that impact PSP include access to knowledge and information (Cooper and Nisbet 2016; Evensen 2017) and perceptions of the living environment (Nielsen and Smyth 2008). Psychological factors, particularly trust, play a crucial role in shaping PSP (Mayer 2016). In general, individuals may adopt specific behaviors based on their perceptions, prior experiences, and sense of complacency (Wang and Kapucu 2006), which in turn yield broader implications for society.

Building on existing research, this study defines public safety perception as a multidimensional construct encompassing individuals' systematic assessments of social order, stability, public services, and social welfare. This definition provides a framework for examining how Internet use may shape perceptions across these domains.

2.2. Impact of internet use on PSP

The Internet has become deeply embedded in daily life (Filsinger and Freitag 2019). As security concerns gain prominence, scholars have increasingly examined its role in shaping these issues (Abubakar and Al-Zyoud 2021; Wang et al. 2022). In public safety governance, the Internet functions as a paradoxical tool, offering both benefits and drawbacks depending on its use (Wang et al. 2022). It lowers transactional costs in crisis communication and accelerates accurate information dissemination through decentralized networks. Empirical evidence highlights this capacity, particularly during the COVID-19 pandemic, when governments used digital infrastructure for real-time epidemiological updates, helping to counter misinformation and strengthen political trust (Lovari 2020). Beyond crisis contexts, the Internet can foster positive outcomes such as social support, identity formation, and belonging (Yue, Zhang, and Xiao 2024), build interpersonal connections (McKenna and Bargh 1998), and therefore enhance life satisfaction (Ryan and Deci 2000).

Nevertheless, many studies find that Internet use can negatively influence PSP. For most people, understanding of risk events depends heavily on available information, with Internet-enabled media becoming a primary source. The online circulation and amplification of risk-related content shape perceptions and influence broader societal views on safety. In Korea, for example, the Internet has been a major channel for disseminating risk narratives rapidly (Chung 2011), while in China, online discourse on food safety has a greater impact on perceptions than personal experiences (Mou and Lin 2014). Research has also documented specific cases where online platforms heightened public concerns: during the COVID-19 pandemic, risk information shared on Twitter significantly increased perceived risk compared with factual information (Hopfer et al. 2021). Negative impacts on PSP resulting from Internet use have caused negative social consequences. These include a reduced willingness to participate in health promotion and disease prevention activities (Rubinelli et al. 2022), as well as an increase in negative attitudes and emotional risks toward political authority (Scardigno and Mininni 2020).

According to the Social Amplification of Risk Framework (SARF), pioneered by Kasperson et al. (1988), social amplification can intensify not only signals about a risk but also perceptions of that risk, related behaviors, and even the risk itself and its consequences (Kasperson et al. 2022). In the digital age, online platforms have become hyper-efficient 'amplification stations', capable of spreading risk information across global communication networks within minutes, fundamentally reshaping the dynamics of risk propagation (Ng, Yang, and Vishwanath 2018). To attract attention, these platforms often release or algorithmically prioritize large volumes of risk-related content—such as reports on riots, terrorism, sabotage, or protests—amplifying their visibility and perceived urgency. As recipients of this amplified information, the public is now more frequently exposed to social risks through Internet media than through traditional channels, which can lead to a sustained decline in PSP. Therefore, we propose the following hypothesis.

H1: Compared to non-use, Internet use significantly weakens the PSP of residents.

2.3. Impact of internet use purposes on PSP

If Internet use significantly affects PSP, does this effect vary according to different purposes of use? Addressing this question requires examining the diverse motivations behind online engagement. A widely accepted premise in media studies is that individuals choose and use mass media or communication channels based on specific purposes, each oriented toward distinct goals; these choices in turn fulfill the users' intended needs (Blumler and Katz 1974). Research on infotainment suggests that social media serves two primary functions: acquiring social information and satisfying entertainment needs. While there are some connections between these two functions (Marinov 2020; Savolainen 2022), they differ substantially in subject matter, format, and style (Brants and Neijens 1998). Guided by this distinction, this study treats information-oriented use and entertainment-oriented use as the two fundamental purposes of Internet activity and proposes hypotheses based on these categories.

On one hand, examining the relationship between public access to social information *via* the Internet and PSP is essential. When individuals seek news online, they frequently encounter fear-inducing content—such as alarming language, disturbing images, or other emotionally charged material—designed to evoke insecurity and lower perceived safety. Fear appeal theory offers useful insight here, suggesting that negative messages intended to elicit fear can effectively persuade individuals to accept the underlying message, particularly when the induced fear is strong enough to motivate action (Keller 1999). Moreover, research shows that negative events exert a stronger influence on psychological states and decision-making than neutral or positive events (Eslami, Ghasemaghsei and Hassanein 2018). As information and communication technology (ICT) advances, negative content on social media tends to draw more attention than positive content (Primack et al. 2019). In contemporary digital environments, however, exposure to such content may be shaped not only by users' active information seeking but also by passive feed consumption, as recommendation systems increasingly structure users' attention and information consumption on social media (Fletcher and Nielsen 2018; Yu et al. 2024). As a result, individuals may encounter safety-related information even when they are not intentionally searching for it, and repeated exposure to such content may intensify perceptions of insecurity. Consequently, as people increasingly turn to the Internet for news, their exposure to negative reporting is likely to grow, potentially contributing to a gradual decline in overall PSP. Based on this reasoning, we propose our second hypothesis.

H2: Internet use for social information significantly decreases the residents' PSP.

On the other hand, the relationship between public access to entertainment content *via* the Internet and PSP also warrants attention. This form of Internet use has drawn growing scholarly interest (Lu, Tong, and Zhu 2020) for its ability to broaden recreational opportunities and heighten the role of leisure in daily life (Qian et al. 2022). According to Uses and Gratifications (U&G) Theory, individuals choose specific media to meet needs for diversion, social interaction, cognitive enrichment, or emotional regulation (Katz, Blumler, and Gurevitch 1974; Quan-Haase and Young 2010), and the Internet provides new, highly accessible ways to fulfill these needs (Whiting and Williams 2013). In entertainment-oriented Internet use—such as streaming videos, gaming, or engaging on social media—users actively seek content aligned with their intrinsic motivations while also satisfying social interaction needs through interactive formats like multiplayer games or live-streaming. Such platforms allow users to form virtual communities, strengthen their sense of belonging and social capital (Dhir and Tsai 2017), and cultivate optimism for future social engagement (Liu et al. 2023). Empirical evidence further suggests that these gratifications from online entertainment (Sun et al. 2023) can divert attention from negative news and shape broader societal attitudes, including PSP (Yi, Han, and Long 2023).

However, a competing perspective suggests that entertainment-oriented Internet use can also operate through a fear appeal mechanism. Many entertainment producers deliberately integrate sensational themes—such as depictions of doomsdays, disasters, and pandemics—to provoke strong emotional responses. While such content may boost platform engagement and profits, it can simultaneously heighten public anxiety and erode perceptions of safety. Moreover,

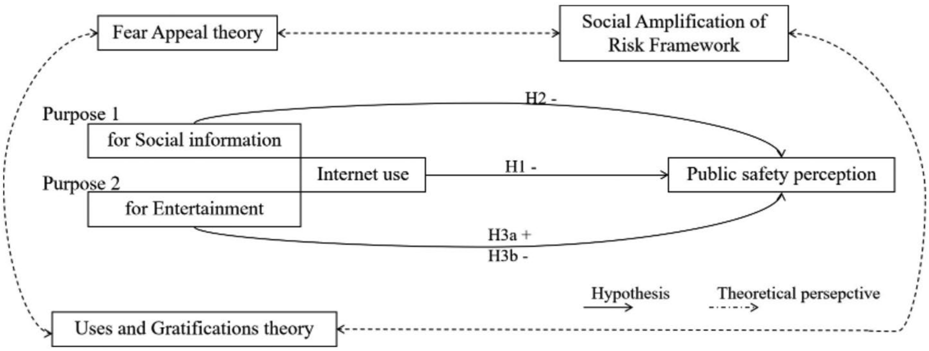

Figure 1. Hypotheses model with theoretical perspectives.

during the COVID-19 pandemic, heightened public awareness of security risks influenced patterns of online entertainment consumption (Chmielarz et al. 2022). This raises the question of whether recreational Internet use exerts different effects on PSP before and after the pandemic. Building on these theoretical insights, this study proposes two competing hypotheses.

H3a: Internet use for entertainment significantly increases the resident' PSP.

H3b: Internet use for entertainment significantly decreases the resident' PSP.

By integrating the theoretical framework with the independent variables related to Internet use, this study constructs a comprehensive hypothetical model. Figure 1 presents the proposed hypotheses, which link Internet use—and its two primary purposes—to PSP. Each independent variable is labeled with the corresponding theoretical perspective that underpins it.

3. Data, variables and methods

3.1. Data source

This study draws on data from the CSS2017 and CSS2021, conducted by the Institute of Sociology at the Chinese Academy of Social Sciences. Launched in 2005, the CSS is a nationwide, biennial, repeated cross-sectional survey that employs probabilistic sampling to cover a new, representative group of 7000–10,000 households across 31 provincial-level administrative regions in mainland China each wave. Due to geographic sensitivities and state policy considerations, publicly available data exclude responses from Xinjiang. This exclusion has minimal impact on representativeness, as Xinjiang's population constitutes only a small share of mainland China's total, and similar exclusions are common in Chinese census data. In 2017, the CSS collected 10,143 valid responses from adults aged 18–71 (5607 females, 4536 males). In 2021, it collected 10,136 valid responses from the same age range (5671 females, 4465 males). After excluding cases with missing data on variables of interest, the final analytical samples consisted of 7906 respondents in 2017 and 7756 respondents in 2021, both of which are adequately representative of the national population.

3.2. Variable design

1. *Dependent Variables*
   PSP is a fundamental human need, encompassing social order, stability, public services, and social welfare—all of which help reduce feelings of insecurity, anxiety, and worry.

To measure this perception, the CSS questionnaires allow respondents to express their subjective feelings, perceptions, and evaluations of public safety. In both the 2017 and 2021 waves, participants were asked to rate their sense of safety across nine domains: personal and family property safety, personal safety, transportation safety, medical care safety, food safety, occupational safety, personal information and privacy safety, environmental safety, and overall societal safety. Responses were recorded on a four-point scale: very unsafe (1), not too safe (2), relatively safe (3), and very safe (4). Higher scores indicated a stronger sense of public safety. Internal consistency and construct validity were assessed using Cronbach's alpha, the KMO index, and Bartlett's test. For CSS2017, Cronbach's alpha was 0.86, the KMO index was 0.90, and Bartlett's test yielded $p < 0.001$. For CSS2021, Cronbach's alpha was 0.88, the KMO index was 0.91, and Bartlett's test also yielded $p < 0.001$. These results indicate a high level of consistency among the nine items in both survey years. Based on this, a composite variable was constructed by averaging the scores of the nine items, representing each respondent's overall PSP.

2. *Independent Variables*

The primary independent variable in this study is Internet usage. Using data from CSS2017 and CSS2021, we measure this variable in two ways. First, we determine whether respondents use the Internet, creating a binary variable—Internet use—coded as 1 for individuals who report using the Internet and 0 for those who do not. Respondents who indicated that they do not use the Internet were excluded from subsequent analyses examining how different purposes of Internet use relate to PSP.

The second independent variable examines the purposes of Internet usage. Drawing on the CSS2017 and CSS2021 questionnaires, we categorized selected items into two groups—social information–oriented usage and entertainment-oriented usage—based on their thematic alignment. For each item, respondents reported usage frequency on a six-point scale (1 = never, 6 = almost every day). The average score across relevant items was calculated to construct the two variables. The specific questionnaire items used for each category are detailed in Table 1.

3. *Control Variables*

Based on prior research (Han, Sun, and Hu 2017; Wu and Li 2017), this study includes a range of control variables: gender, age, education, marital status, political affiliation, household registration status (hukou), religious beliefs, personal annual income, and geographic region. Gender, marital status, party affiliation, hukou, and religion were coded as dummy variables, with values of 0 or 1 according to standard conventions (e.g. 0 = female, 1 = male; 0 = non-CPC member, 1 = CPC member). Age was recorded in years, and personal annual income was log-transformed. Education was recoded into six ordered

Table 1. Original items of CSS questionnaires for internet use proposes.

	CSS2017	CSS2021
Social information oriented usage	Item 1: browsing political news online Item 2: browsing amusement news Item 3: searching information	Item 1: browsing current affair news online (such as: political news)
	Item 4: chatting and making friends	Item 3: chatting and making friends (such as chatting *via* WeChat or QQ)
	Item 5: participating in or forwarding topics	Item 5: learning and education
Entertainment oriented usage	Item 6: playing games online Item 9: listening to music, watching video, or reading novels online	Item 2: Entertainment and leisure (such as: playing games online, listening to music, watching video, or reading novels online)
	Item 7: shopping or making payments online	Item 6: online shopping and life services (such as online shopping, food delivery, navigation, and location services)

categories, from 1 (uneducated) to 6 (graduate degree). Geographic region followed the classification of the National Bureau of Statistics,[1] with provinces in the eastern, central, and western regions coded as 1, 2, and 3, respectively.

3.3. Methods

This study examines the relationship between Internet use and PSP, measured as a continuous variable, using ordinary least squares (OLS) regression as the primary method. The main independent variables are Internet use (binary) and two composite measures capturing its purposes (social information and entertainment), with demographic and socioeconomic controls included. Multicollinearity diagnostics indicated no serious concern: condition numbers were below 30 and all VIFs were below 2.2 in both waves. Additionally, multilevel modeling is used to account for the nested data structure (individuals within provinces) and capture provincial-level variation in the effects of Internet use. To verify robustness, two-stage least squares (2SLS) regression with instrumental variables is applied to address potential endogeneity. Furthermore, ordered logistic regression (Ologit) was employed as a supplementary test to further verify the robustness of the OLS model and its results.

4. Research findings

4.1. Descriptive statistics

Table 2 presents the descriptive statistics of the key variables. The results indicate that the PSP is relatively high in both waves, with mean values consistently above 2 on the 1–4 scale. Meanwhile, Internet use showed a noticeable increase: the average frequency of usage rose by approximately 0.3 points between 2017 and 2021, reflecting the continued integration of the Internet into daily life during this period.

4.2. The relationship between Internet use and PSP

Table 3 reports the results of both the OLS and ordered logit (Ologit) models that examine the impact of Internet use on PSP while controlling for individual- and household-level factors. Across Models 1 and 2 (OLS), Internet use consistently shows a significant negative effect on PSP. Compared to non-users, individuals who use the Internet report lower levels of PSP, even after accounting for sociodemographic and attitudinal variables. This negative relationship is

Table 2. Descriptive statistic results of variables.

Variable	*N*(2017)	Mean	Std. Dev.	Min	Max	*N*(2021)	Mean	Std. Dev.	Min	Max
PSP	7906	2.863	0.493	1	4	7756	3.111	0.456	1	4
Internet use	7906	0.437	0.496	0	1	7756	0.730	0.444	0	1
Social information oriented internet use	3401	4.046	1.135	1	6	5663	3.388	1.263	1	6
Entertainment oriented internet use	3401	2.933	1.093	1	6	5663	2.839	1.255	1	6
Gender	7906	0.463	0.499	0	1	7756	0.466	0.499	0	1
Age	7906	45.329	14.160	17	70	7756	44.960	14.295	18	70
Education	7906	3.121	1.218	1	6	7756	3.344	1.222	1	6
Marital status	7906	0.871	0.335	0	1	7756	0.849	0.358	0	1
Political Status	7906	0.102	0.303	0	1	7756	0.110	0.313	0	1
Hukou	7906	0.315	0.464	0	1	7756	0.358	0.479	0	1
Religious beliefs	7906	0.129	0.335	0	1	7756	0.137	0.344	0	1
Income (log)	7906	7.907	3.855	0	14.221	7756	8.402	3.828	0	14.221
Area	7906	1.808	0.833	1	3	7756	1.877	0.828	1	3

Table 3. Analysis of relationship between internet use and PSP.

	(1)CSS2017	(2)CSS2021	(3)CSS2017	(4)CSS2021
PSP	OLS	OLS	Ologit	Ologit
Internet use	−0.087***	−0.086***	−0.378***	−0.342***
	(0.014)	(0.014)	(0.053)	(0.054)
Gender	0.042***	0.028***	0.182***	0.143***
	(0.012)	(0.011)	(0.042)	(0.042)
Age	0.004***	0.001**	0.014***	0.005**
	(0.001)	(0.001)	(0.002)	(0.002)
Education	−0.037***	−0.051***	−0.149***	−0.216***
	(0.006)	(0.006)	(0.023)	(0.023)
Marital status	−0.080***	−0.073***	−0.325***	−0.260***
	(0.020)	(0.018)	(0.072)	(0.071)
Political status	0.040*	0.119***	0.125*	0.409***
	(0.019)	(0.017)	(0.068)	(0.068)
Hukou	−0.071***	−0.032***	−0.271***	−0.132***
	(0.013)	(0.012)	(0.048)	(0.046)
Religious beliefs	−0.006	0.005	−0.058	0.004
	(0.016)	(0.015)	(0.059)	(0.059)
Income	0.001	0.001	0.007	0.001
	(0.002)	(0.001)	(0.006)	(0.006)
Cons	2.899***	3.323***		
	(0.036)	(0.035)		
R^2	0.061	0.045		
Observations	7,906	7,756	7,906	7,756

Standard errors in parentheses, ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

not only statistically significant but also robust across both 2017 and 2021 datasets. Models 3 and 4, which employ Ologit estimation, further corroborate these results: Internet users are significantly less likely to report higher categories of perceived public safety compared to their non-user counterparts. Together, these findings provide strong evidence that Internet use is associated with diminished perceptions of safety. Despite the expansion of Internet penetration and its growing role in daily life, individuals engaging online tend to experience lower public safety perception. Hypothesis 1 is therefore supported.

Among the control variables, gender, age, and political status are positively associated with PSP. Men generally report higher levels of PSP than women, while older respondents also tend to feel safer, though the effect of age is relatively modest. Political affiliation exerts a stronger influence, with CPC members expressing higher levels of PSP than non-members, reflecting the growing salience of political identity in shaping security perceptions. By contrast, education is negatively associated with safety perceptions, suggesting that more educated individuals are more critical and sensitive to public security issues, consistent with prior research highlighting their lower tolerance of insecurity (Hu, Zeng, and Yi 2019). Marital status and hukou also show significant negative associations: married individuals and urban residents perceive lower levels of public safety compared to unmarried and rural counterparts. Finally, religious beliefs and personal income do not show significant correlations, indicating that these factors are less relevant in explaining PSP.

4.3. The relationship between purposes of internet use and PSP

The study examines how different purposes of Internet use—social information-oriented and entertainment-oriented—relate to PSP. The results reported in Table 4 show that in 2017, only social information-oriented Internet use was significantly associated with lower PSP, while entertainment-oriented use had no significant effect. By 2021, however, both types of Internet use displayed significant correlations with PSP: social information-oriented use continued to exert a negative influence, though its effect weakened compared to 2017, whereas entertainment-oriented use showed a significant positive association. This cross-wave difference

Table 4. Analysis of relationship between purposes of internet use and PSP.

	(1) CSS2017	(2) CSS2017	(3) CSS2017	(4) CSS2021	(5) CSS2021	(6) CSS2021
PSP			OLS			
Soci-purpose	−0.016**		−0.016**	−0.011*		−0.016***
	(0.007)		(0.008)	(0.005)		(0.006)
Enter-purpose		−0.006	−0.000		0.017***	0.021***
		(0.007)	(0.007)		(0.005)	(0.005)
Gender	0.058***	0.059***	0.058***	0.054***	0.050***	0.048***
	(0.016)	(0.016)	(0.016)	(0.012)	(0.012)	(0.012)
Age	0.002**	0.002**	0.002*	0.001	−0.000	0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Education	−0.021**	−0.025***	−0.021**	−0.048***	−0.038***	−0.044***
	(0.009)	(0.009)	(0.009)	(0.007)	(0.007)	(0.007)
Marital stat	−0.070***	−0.070***	−0.070***	−0.045**	−0.042**	−0.041**
	(0.023)	(0.023)	(0.023)	(0.019)	(0.019)	(0.019)
Political stat	0.060**	0.057**	0.060**	0.102***	0.104***	0.095***
	(0.023)	(0.023)	(0.024)	(0.019)	(0.019)	(0.019)
Hukou	−0.023	−0.023	−0.023	−0.017	−0.012	−0.014
	(0.018)	(0.018)	(0.018)	(0.013)	(0.013)	(0.013)
Religious beliefs	−0.015	−0.016	−0.015	−0.003	−0.003	−0.004
	(0.023)	(0.023)	(0.023)	(0.017)	(0.017)	(0.017)
Income	0.002	0.002	0.002	0.001	0.001	0.001
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Cons	2.850***	2.814***	2.850***	3.243***	3.158***	3.208***
	(0.056)	(0.057)	(0.060)	(0.042)	(0.039)	(0.043)
R^2	0.017	0.016	0.017	0.021	0.022	0.023
Observations	3,401	3,401	3,401	5,663	5,663	5,663

Standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

suggests that while consuming social information online is negatively associated with PSP in both waves, the estimated association is weaker in 2021 than in 2017. In contrast, the growing positive role of entertainment-oriented use indicates that leisure and recreational Internet activities may have begun to buffer anxieties and enhance feelings of security. Together, these findings support Hypotheses 2 and 3a and point to the need for closer examination of why entertainment-oriented Internet use exhibited a cross-wave divergence from a null effect in 2017 to a positive association in 2021.

Several factors may help explain this cross-sectional difference. From the perspective of Mood Management Theory, entertainment-oriented Internet use allows individuals to select content that reduces negative affect, diverts attention away from threat-related information, and helps regulate mood (Zillmann 2000). In the pandemic-era context of 2021, when digital media became an important resource for coping with stress, anxiety, and social disruption, online entertainment may have functioned less as a risk amplifier than as a coping resource that buffered stress and restored a sense of normalcy and control (Eden et al. 2020; Wolfers et al. 2024). In this sense, digital media can be understood as a form of digital resilience: not merely a channel of exposure, but a temporary 'life-support system' that provided escapism, emotional relief, and practical continuity during a period of heightened uncertainty. These buffering mechanisms may help explain why entertainment-oriented Internet use was not significantly associated with PSP in 2017 but was positively associated with PSP in the 2021 wave.

4.4. Heterogeneous and contextual effects

To further probe heterogeneity in the relationship between Internet use and PSP, subgroup analyses were conducted by gender, age, and educational attainment (see Appendix A for detailed regression tables). The results reveal notable variations. First, women consistently reported lower PSP than men when using the Internet. This gap may reflect gendered differences in exposure to, and interpretation of, security-related information online, as well as women's heightened sensitivity to safety issues in both online and offline contexts. Second, across all age groups, Internet use was associated with diminished PSP. However, the magnitude of this effect manifested differently across the two survey waves. For younger respondents (aged 18–29), the negative association was especially pronounced in 2017 but was statistically insignificant in the 2021 wave. One possible explanation is that younger individuals, being generally healthier and less vulnerable to risks such as COVID-19, may have been less influenced by panic-inducing content in the later wave of the survey. Finally, differences by education level were also apparent. While the effect of Internet use on PSP was strongly negative among those without college education, it was weaker and statistically insignificant among college-educated respondents in 2021. This pattern suggests that higher levels of education may equip individuals with greater critical thinking skills, enabling them to filter or contextualize online information and thereby mitigate its negative impact on public safety perception.

Additionally, this study also distinguishes different types of PSP. Individual-level perceptions include concerns about personal and family property, personal safety, occupational safety, and privacy, while societal-level perceptions cover issues such as transportation, medical care, food, environmental safety, and overall social order. Separate regression analyses (see Appendix B) show that Internet use significantly undermines both types of perceptions, but the effect is stronger at the societal level. A useful way to interpret this pattern is through the impersonal impact hypothesis: digital media expose users to vivid reports of crime, accidents, food safety scandals, environmental hazards, and other forms of distant suffering, making problems in the broader social world psychologically salient even when users' immediate daily lives remain relatively stable (Moy 2008). As a result, Internet exposure may disproportionately shape judgments about societal safety, which rely more heavily on mediated information, while exerting a weaker influence on personal safety assessments, which remain more anchored in direct local experience. This asymmetry is also consistent with Third-Person Effect and Optimistic Bias arguments, as individuals may perceive media-driven risks as affecting others more than themselves and may believe that they are personally less vulnerable to negative events (Gunther and Mundy 1993). Further analyses of Internet use purposes reinforce this pattern. Social information–oriented use consistently correlates with more negative perceptions, particularly at the societal level, as frequent exposure to critical news and discussions amplifies concerns about systemic problems. In contrast, entertainment-oriented use shows divergent effects: while it had little impact in 2017, by 2021 it was positively associated with both individual and societal safety perceptions. This divergence across periods can be explained by the growing role of online entertainment platforms, which during the COVID-19 pandemic became central to daily life and may have functioned less as channels of risk amplification than as coping resources that provided escapism, emotional relief, and a greater sense of normalcy and control, thereby enhancing users' sense of security.

Beyond subgroup and perception-type differences, we also tested whether provincial contexts moderated the Internet–safety relationship. To account for the hierarchical structure of the data—individuals nested within provinces—this study employed multilevel modeling to complement the individual-level regressions (see Appendix C for full results). Multilevel models allow intercepts and slopes of Internet use to vary across provinces, thereby capturing contextual differences that cannot be addressed by standard OLS or Ologit models. The results reveal that while Internet use consistently shows a negative association with PSP, a portion of the variation in these perceptions is explained by provincial-level factors. In 2017, between-province variance was negligible, suggesting that Internet use exerted a broadly similar effect nationwide. By 2021, however, provincial differences were more pronounced, with approximately 3% of the variance in public safety perception occurring at the provincial level. Notably, in the 2021 multilevel models, higher provincial education levels and greater government network transparency were both associated with lower average PSP, whereas neither contextual factor was significant in 2017. At the same time, the government network transparency index did not significantly moderate the Internet effect, suggesting that it mattered more for baseline levels of PSP than for systematically buffering the negative association between Internet use and PSP. The multilevel findings therefore confirm the robustness of the main models while also suggesting that Internet use interacts with contextual features, amplifying regional heterogeneity in perceptions of safety during the COVID-19 period.

5. Causal inference through 2SLS estimation

Internet use and PSP may be jointly determined: online activity can shape public safety perception, while concern about safety can itself prompt more online search and communication. To address this endogeneity, we implement 2SLS with two instruments: (i) the log of last-year telecommunications expenditure (Tele-cost), which is strongly related to connectivity intensity and plan choice yet plausibly unrelated to current PSP except through Internet use; and (ii) participation in social group activities (Tuan-ti).

Tele-cost is a relevant instrument because higher telecommunications spending is closely associated with the affordability and intensity of digital connectivity, making it a strong predictor of whether and how frequently respondents use the Internet. Conditional on rich individual covariates, however, telecommunications expenditure is unlikely to affect PSP directly; its most plausible effect operates through increased online access and exposure. Tuan-ti is also a relevant instrument because participation in social organizations creates greater opportunities and incentives for online communication, information exchange, and coordination, thereby increasing Internet use. Once individual socioeconomic characteristics are taken into account, its remaining direct association with PSP is less plausible, making Internet-mediated communication and information acquisition the more credible pathway. Empirically, first-stage diagnostics indicate that the instruments are strongly associated with Internet use, and, in the overidentified specifications, Hansen J tests fail to reject the overidentifying restrictions, providing additional reassurance about instrument validity. To further reduce concerns about alternative channels, we estimate specifications with additional provincial covariates—including GDP per capita, crime rate, Internet penetration, urbanization, unemployment, and average education—and province fixed effects as robustness checks. Under these more demanding specifications, the IV estimates remain substantively similar, which strengthens confidence in the plausibility of the exclusion restriction.

Table 5 reports the 2SLS results (first stages in columns (1), (3), (5); second stages in columns (2), (4), (6)). In 2017 (Panel A), the Cragg–Donald Wald F and Sanderson–Windmeijer multivariate F statistics are 58.78 and 65.59 for Internet use; 40.72 and 43.29 for social-information use; and 18.01 and 15.33 for entertainment use—well above the conventional weak-IV thresholds. In 2021 (Panel B), the corresponding F's are even larger (73.52/92.10 for Internet use; 70.57/78.87 for social-information; 34.42/38.59 for entertainment). Hansen J tests (p-values 0.16–0.78) fail to reject joint exogeneity of the instruments. Although Tele-cost is not individually significant for the social-information first stage in 2021, the joint F with Tuan-ti (≈70.6) indicates strong combined relevance.

Panel A indicates that in 2017, the instrumental variable results provide evidence of a causal relationship wherein Internet use reduces PSP, corroborating the baseline regressions. When disaggregating by purpose, social-information use has a negative effect on PSP, and entertainment use is also negative. The latter contrasts with the OLS null in 2017, suggesting that endogeneity—e.g. safer-feeling individuals engaging more in online leisure (reverse causality) or measurement error—masked an underlying adverse effect. In 2017s pre-pandemic environment, entertainment content likely included sensational or violent themes and substituted for offline social buffering, which, once instrumented, reveals a net reduction in PSP.

Panel B shows that in 2021, the average effect of Internet use is also negative. However, purpose-specific effects diverge: social-information use turns positive, and entertainment use is clearly positive. This pattern is consistent with pandemic-era dynamics: exogenous increases in connectivity and group participation (captured by Tele-cost and Tuan-ti) channel users toward

Table 5. Analysis of the 2SLS estimation.

	Panel A					
	(1) CSS2017	(2) CSS2017	(3) CSS2017	(4) CSS2017	(5) CSS2017	(6) CSS2017
Variables	Internet use	PSP	Soci-purpose	PSP	Enter-purpose	PSP
Internet use		−0.548***				
		(0.125)				
Soci-purpose				−0.101**		
				(0.045)		
Enter-purpose						−0.157**
						(0.070)
Tuan-ti	0.096***		0.317***		0.121***	
	(0.012)		(0.037)		(0.040)	
Tele-cost	0.022***		0.059***		0.093***	
	(0.003)		(0.019)		(0.021)	
Control Var	YES	YES	YES	YES	YES	YES
R^2	0.443	–	0.228	–	0.268	–
Cragg-Donald Wald F	58.778***		40.724***		18.013***	
Sanderson-Windmeijer multivariate F	65.59***		43.286***		15.33***	
Hansen J	1.987		0.877		0.121	
(p value)	0.159		0.349		0.729	
Observations	7,906		3,401			
	Panel B					
	(1) CSS2021	(2) CSS2021	(3) CSS2021	(4) CSS2021	(5) CSS2021	(6) CSS2021
Variables	Internet use	PSP	Soci-purpose	PSP	Enter-purpose	PSP
Internet Use		−0.338***				
		(0.102)				
Soci-purpose				0.064*		
				(0.034)		
Enter-purpose						0.097**
						(0.047)
Tuan-ti	0.046***		−0.293***		−0.192***	
	(0.004)		(0.024)		(0.024)	
Tele-cost	0.030***		−0.022		−0.050***	
	(0.004)		(0.014)		(0.016)	
Control Var	YES	YES	YES	YES	YES	YES
R^2	0.322	0.003	0.314	–	0.287	0.001
Cragg-Donald Wald F	73.518***		70.569***		34.415***	
Sanderson-Windmeijer multivariate F	92.103***		78.87***		38.590***	
Hansen J	0.36		0.663		0.080	
(p value)	0.549		0.415		0.777	
Observations	7,756		5,663			

Standard errors in parentheses, ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

authoritative updates, public-health guidance, and supportive online communities, which mitigate fear and improve PSP. In short, once selection into information seeking is instrumented, the causal effect of 2021 social-information use appears mildly reassuring rather than alarming.

Across both waves, the 2SLS evidence suggests a causal link between Internet use and PSP while revealing time- and purpose-specific heterogeneity: negative overall effects persist, but the composition of online activity matters—especially in the 2021 wave, when entertainment and (to a lesser extent) social-information use were estimated to have positive effects. These results strengthen the paper's identification strategy, support H1 (overall negative association), refine H2/H3 (purpose-specific effects), and underscore the policy relevance of risk communication quality and supportive online environments.

6. Conclusion and discussion

This study advances understanding of how Internet use shapes PSP in China. Using nationally representative data from 2017 and 2021, we find a robust negative average association between

Internet use and PSP across OLS and Ologit models, and we point toward a potential causal link with 2SLS estimates. The effect is heterogeneous. By purpose, social-information use (news browsing, information seeking, and online discussion) is consistently associated with lower PSP, whereas entertainment-oriented use is positively associated with PSP in the 2021 wave. By perception type, the Internet depresses societal-level safety judgments more than individual-level assessments, indicating that online exposure more powerfully shapes evaluations of collective order and governance than everyday personal security. Subgroup analyses show stronger negative effects for women and the less educated, and multilevel models reveal that provincial context was more salient in 2021 (with higher provincial average education level predicting lower PSP), suggesting that the Internet–PSP link is filtered by social environments. Together, these findings depict an information ecosystem in which the composition of use and context matter as much as overall exposure.

Building on multiple theoretical perspectives, the relationship between Internet use and PSP unfolds through both risk-amplifying and risk-buffering mechanisms. Because negative events exert stronger psychological and behavioral effects than neutral or positive ones, online information producers—guided by fear appeal theory—often craft panic-inducing messages to capture attention. Within the SARF, today's digital platforms act as hyper-efficient 'amplification stations', rapidly magnifying and disseminating such content, which in turn lowers PSP; this is particularly pronounced for socio-informational use, where frequent exposure to alarming news fosters heightened insecurity. In contrast, U&G theory and Mood Management Theory jointly suggest that entertainment-oriented Internet use may serve both gratification-seeking and mood-regulation functions, helping individuals reduce negative affect and divert attention away from threat-related information. In the pandemic-era context, such use may have functioned less as a risk amplifier than as a coping resource that buffered stress and restored a sense of normalcy and control. In this sense, digital media could temporarily serve as a form of digital resilience—a 'life-support system' that provided escapism, emotional relief, and continuity during a period of heightened uncertainty. Consequently, although its effect on PSP was negligible in 2017, entertainment-oriented Internet use demonstrated a clear positive association by 2021, revealing a nuanced causal story in which different Internet use purposes shape PSP in divergent ways.

This study contributes to the literature on political communication and risk by systematically evaluating how Internet use shapes PSP in China, an important context where online media plays a central role in information dissemination. By disaggregating Internet use into distinct purposes, the analysis reconciles mixed findings in prior research, showing that risk-amplifying and risk-buffering mechanisms can coexist depending on use patterns. Moreover, nuanced evaluations reveal that impact of Internet use on PSP varies across different demographics, level of judgments (societal vs. personal), and regional information ecologies. Methodologically, the paper combines OLS, instrumental-variable estimation, and multilevel modeling to strengthen causal inference and capture heterogeneous and contextual effects. At the same time, the analysis points to directions for future work: stronger designs such as panel data or field experiments, more direct measures that distinguish active information seeking from passive feed consumption and capture algorithmic curation more explicitly, and comparative studies beyond China would deepen understanding of both mechanisms and generalizability.

In light of these findings, this research carries important implications for policymakers, particularly in developing countries where Internet infrastructure is rapidly expanding and netizen populations are growing at unprecedented rates. The Chinese case shows that while digital connectivity can enhance governance capacities and broaden access to information, it also reshapes PSP in complex ways, amplifying insecurities when risk-oriented content dominates and buffering them when entertainment or supportive communities prevail. For governments seeking to harness digital transformation without eroding social stability, this means that investments in infrastructure must be matched with careful attention to the quality of online content, the regulation of misinformation, and the cultivation of trustworthy digital communication channels. Moreover, the heterogeneity of effects across gender, education, and regional contexts underscores the need for tailored communication strategies that address vulnerable groups most susceptible to risk amplification. For developing countries at the threshold of digital expansion, China's experience offers a critical lesson: expanding Internet access is not merely a technological or economic project, but a sociopolitical one that requires proactive strategies to balance risk communication, promote digital literacy, and build resilient PSP in the digital age.

Note

1. The area is classified by China's National Bureau of Statistics, http://www.stats.gov.cn/(accessed on 7 May 2023).

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Data availability statement

The data that support the findings of this study are openly available in Chinese Social Survey at http://sociology.cssn.cn/css_sy/

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